INCHUL KIM (Incheol Kim)

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APPOINTMENTS

2022–Present	**Visual Computing Lab**, KAIST, South Korea, PhD Research Assistant
	– Research on various topics in computer vision and graphics
2021–2022	PTERS company (currently ALOGIC), South Korea, Software Engineer
	– Front-end and back-end web development
2017–2018	**Graphics and Imaging Lab**, Universidad de Zaragoza, Spain, Research Assistant
	– Research on mitigating visual discomfort in 360° video playback on HMDs
2010–2012	Republic of Korea Air Force (compulsory military service)

EDUCATION

2022–Present	**KAIST, South Korea, PhD Candidate in Computer Science**
	– Supervisor: **Prof. Min H. Kim**
2015–2017	**KAIST, South Korea, MSc in Computer Science**
	– Supervisor: **Prof. Min H. Kim**
	– Thesis: Dehazing using Non-Local Regularization with Iso-Depth Neighbor-Fields
2009–2015	**Hanyang University, South Korea, BSc in Computer Science**
	– Summa Cum Laude

AWARDS/SCHOLARSHIPS/FUNDING

- **Best Paper Award (Günter Enderle Award for the Best Paper) [link], Eurographics, 2026**
  – Polychromatic modelling of X-ray physics for CBCT metal artifact reduction.
- **Outstanding Teaching Assistant Award [link (in Korean)], KAIST, 2023**
- National Scholarship: Full Tuition for Graduate Study, Korean Government (2022–)
- **KAIST Breakthroughs [link], KAIST, 2017**
  – Compact hyperspectral imaging at low cost (presented at **ACM SIGGRAPH Asia 2017 [link]**).
- National Scholarship: Full Tuition for Graduate Study, Korean Government (2015–2017)
- National Science and Engineering Undergraduate Scholarship, Korean Government (2013–2014)

ACADEMIC SERVICE

Reviewer:

- IEEE Computer Vision and Pattern Recognition (**CVPR**) 2024, 2025, 2026
- IEEE Transactions on Image Processing (**TIP**) 2024
- IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**) 2024, 2025, 2026
- IEEE Transactions on Visualization and Computer Graphics (**TVCG**) 2024
- Optics Express (**OE**) 2024

Projects

2024–Present	**Hyperspectral Imaging System**, National Research Foundation (NRF), South Korea – Developing a real-time hyperspectral video capture and reconstruction system.
2022–2024	**Intra-oral Scanner**, Dentium, South Korea – Research on a robust, real-time 3D reconstruction using an RGBD camera
2016–2017	**High Dynamic Range Video**, Electronics and Telecommunications Research Institute (ETRI), South Korea – Research on a next-generation codec for high dynamic range video compression

Publications

Refereed International Journals:

[J1] Kiseok Choi, **Inchul Kim**, Jaemin Cho, Hyeongjun Cho, and Min H. Kim (2026), "**Splat-based Metal Artifact Reduction in Cone-Beam CT via Polychromatic Modeling**," *Computer Graphics Forum (**CGF**) (presented at **Eurographics** 2026), 2026.* **[project] [pdf]**

[J2] Ana Serrano, **Incheol Kim**, Zhili Chen, Stephen DiVerdi, Diego Gutierrez, Aaron Hertzmann, and Belén Masiá (2019), "**Motion Parallax for 360° RGBD Video**," *IEEE Transactions on Visualization and Computer Graphics (**TVCG**), 54(5):1817–1827, 2019.* **[project] [pdf]**

[J3] Seung-Hwan Baek, **Incheol Kim**, Diego Gutierrez, and Min H. Kim (2017), "**Compact Single-Shot Hyperspectral Imaging Using a Prism**," *ACM Transactions on Graphics (**TOG**) (presented at **SIGGRAPH Asia** 2017), 36(6):217:1–12, 2017.* **[project] [pdf]**

Peer-reviewed International Conferences:

[C1] Kiseok Choi, Jaemin Cho, **Inchul Kim**, and Min H. Kim (2026), "**Splat-Based Metal Artifact Reduction in Cone-Beam CT via Compact Attenuation Modeling**," *In IEEE Computer Vision and Pattern Recognition (**CVPR**) 2026, Denver, Colorado, Jun. 3–7, 2026.*

[C2] Kiseok Choi, Hyeongjun Cho, **Inchul Kim**, and Min H. Kim (2026), "**Revisiting Pose Sensitivity in Splat-based Computed Tomography under Sparse-view Reconstruction**," *In IEEE Computer Vision and Pattern Recognition (**CVPR**) 2026, Denver, Colorado, Jun. 3–7, 2026.*

[C3] Kiseok Choi, **Inchul Kim**, Dongyoung Choi, Julio Marco, Diego Gutierrez, and Min H. Kim (2023), "**Self-Calibrating, Fully Differentiable NLOS Inverse Rendering**," *In Proceedings of ACM **SIGGRAPH Asia** 2023, Sydney, Australia, Dec. 12–15, 2023.* **[project] [pdf]**

[C4] Donggun Kim, Hyeonjoong Jang, **Inchul Kim**, and Min H. Kim (2023), "**Spatio-Focal Bidirectional Disparity Estimation from a Dual-Pixel Image**," *In IEEE Computer Vision and Pattern Recognition (**CVPR**) 2023, Vancouver, Canada, Jun. 18–22, 2023.* **[project] [pdf]**

[C5] **Incheol Kim** and Min H. Kim (2019), "**Non-local Haze Propagation with an Iso-Depth Prior**," *In Computer Vision, Imaging and Computer Graphics – Theory and Applications, pp 213–238, 2019.* **[project] [pdf]**

[C6] **Incheol Kim** and Min H. Kim (2017), "**Dehazing using Non-local Regularization with Iso-depth Neighbor-Fields**," *In International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications (VISIGRAPP 2017) – Volume 4: VISAPP, pages 77–88, 2017.* **[project] [pdf]**

Patents:

[P1] Min Hyuk Kim, Seung-Hwan Baek, and **Incheol Kim** (2019), "**Method for reconstructing hyperspectral image using prism and system therefor**," ***US Patent**: US20190096044A1, published in Mar. 28, 2019.* **[link]**

[P2] Min Hyuk Kim, Seung-Hwan Baek, and **Incheol Kim** (2019), "**Method and system for reconstructing hyperspectral image by using prism**," ***PCT Patent**: WO2019059632A1, published in Mar. 28, 2019.* **[link]**

[P3] Min Hyuk Kim, Seung-Hwan Baek, and **Incheol Kim** (2019), "**Method for reconstructing hyperspectral image using prism and system therefor**," ***EU Patent****: EP3460427A1, published in Mar. 27, 2019.* **[link]**

REFERENCES

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